

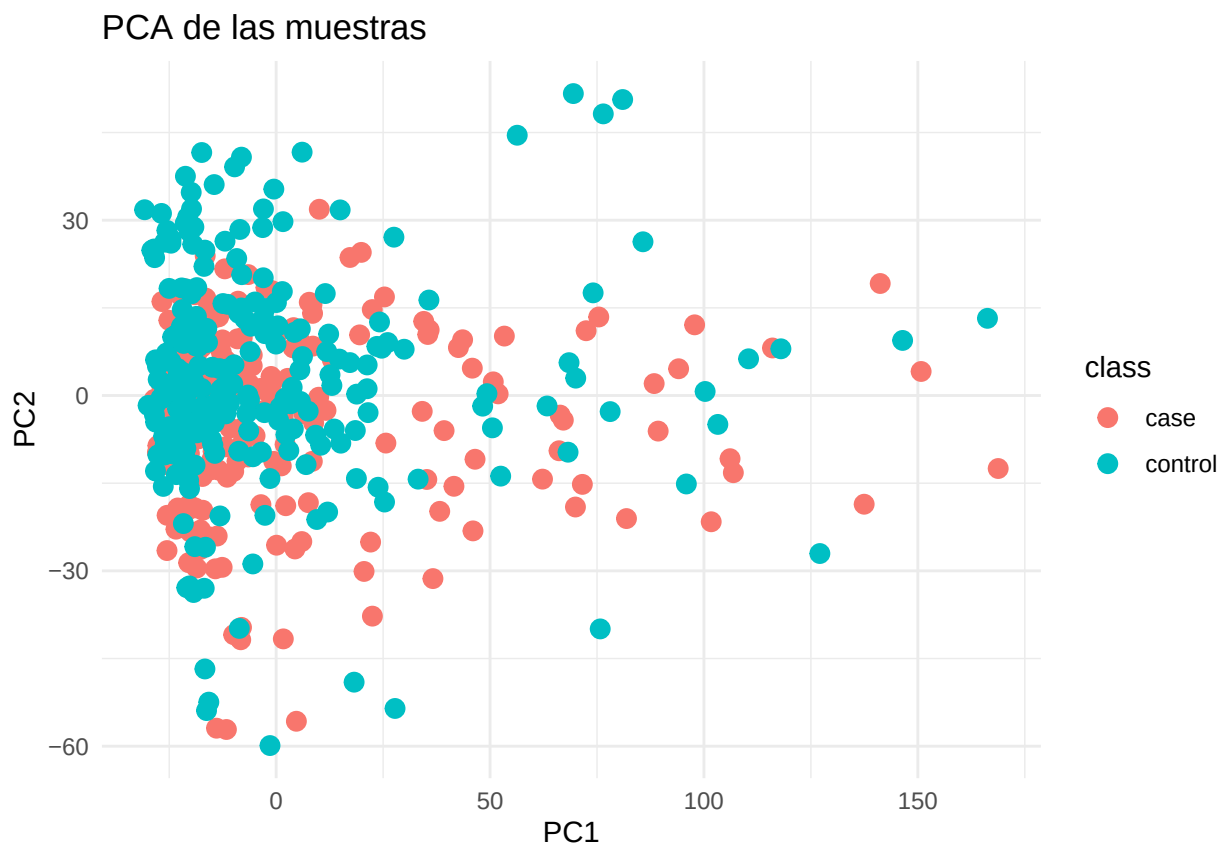
Análisis Exploratorio de Expresión Génica — GSE152075

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Contents

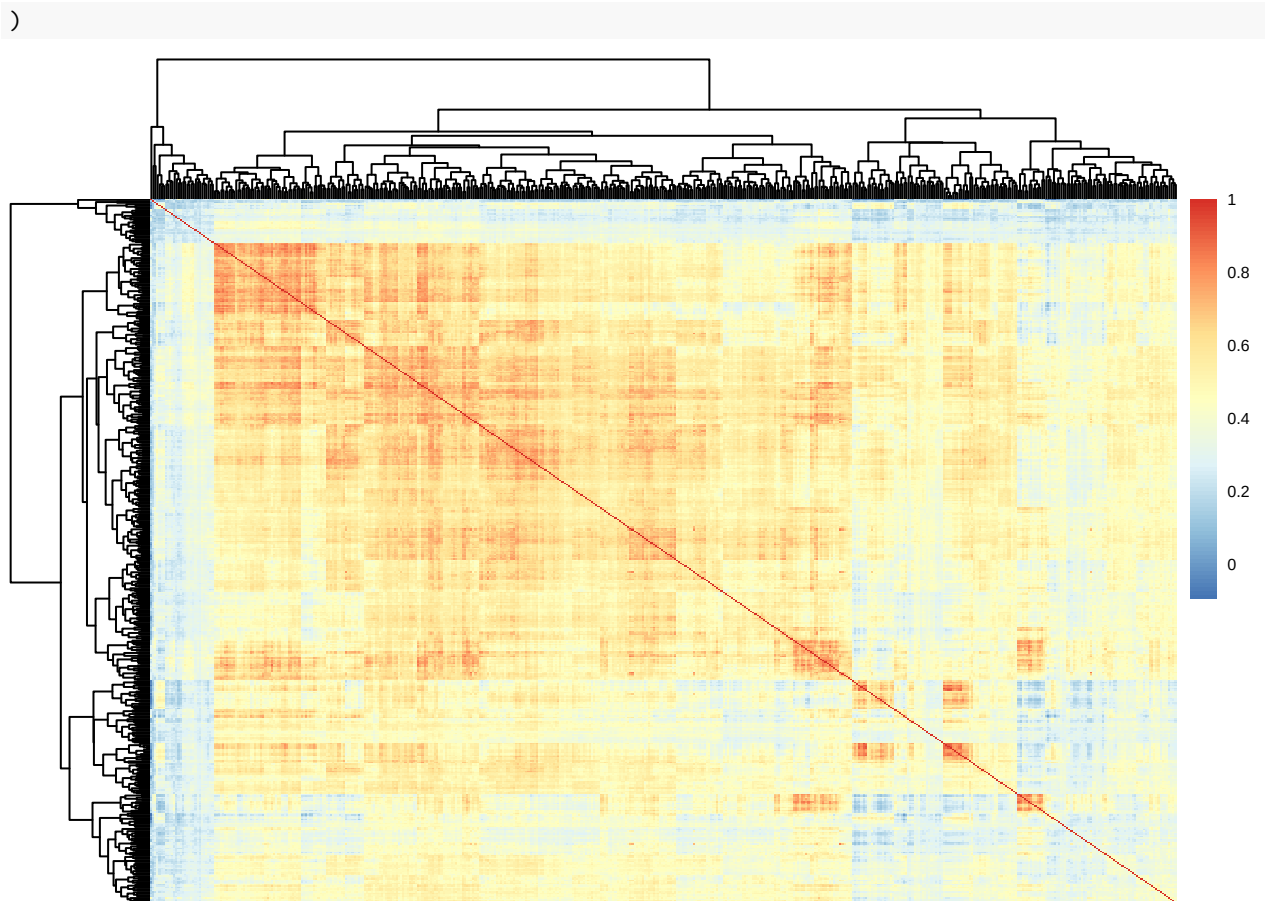
```
pca <- prcomp(t(lcpm), scale. = TRUE)
pca_df <- data.frame(PC1 = pca$x[,1], PC2 = pca$x[,2], class = sample_info$class)

ggplot(pca_df, aes(PC1, PC2, color = class)) +
  geom_point(size = 3) +
  theme_minimal() +
  ggtitle("PCA de las muestras")
```



```
cor_mat <- cor(lcpm)

pheatmap(
  cor_mat,
  show_rownames = FALSE,
  show_colnames = FALSE,
  fontsize = 6)
```



```
group <- sample_info$class

res <- apply(lcpm, 1, function(x) {
  g1 <- x[group == "case"]
  g2 <- x[group == "control"]
  if(length(g1) < 2 | length(g2) < 2) return(c(stat = NA, p.value = NA))
  test <- t.test(g1, g2)
  c(stat = test$statistic, p.value = test$p.value)
})

res_df <- as.data.frame(t(res))
res_df$gene <- rownames(res_df)
res_df$adj.p <- p.adjust(res_df$p.value, method = "BH")
res_df <- res_df[order(res_df$adj.p), ]

# Guardar resultados
write.csv(res_df, "ttest_results_full.csv", row.names = FALSE)
head(res_df, 10)
```

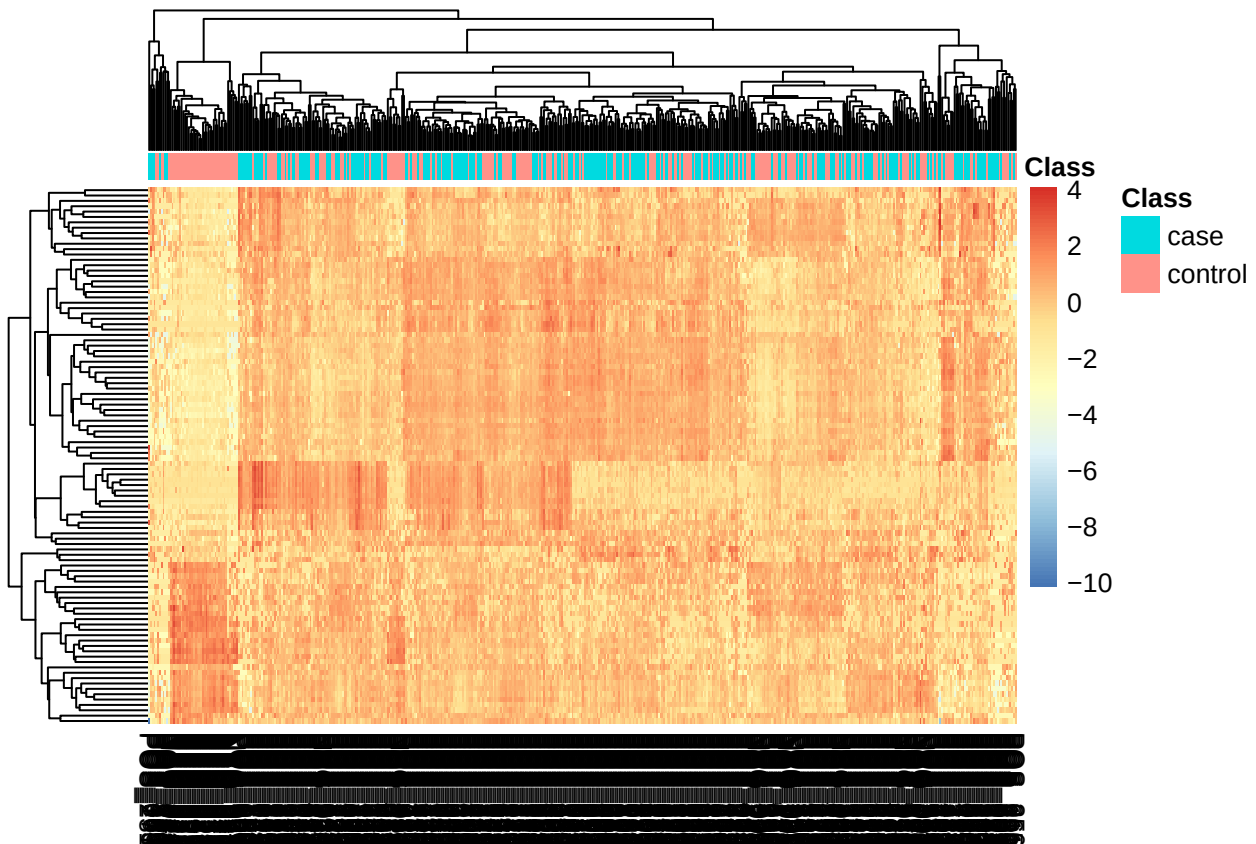
```
##      stat.t      p.value  gene      adj.p
## 22690 6.162758 1.721071e-09 22690 3.712192e-06
## 24108 6.131134 1.820892e-09 24108 3.712192e-06
## 29090 6.309129 6.448969e-10 29090 3.712192e-06
## 34603 5.922950 7.057910e-09 34603 1.079154e-05
## 27432 5.736488 1.709271e-08 27432 1.742317e-05
```

```
## 33109 5.785079 1.537602e-08 33109 1.742317e-05
## 19609 5.706391 2.073348e-08 19609 1.811514e-05
## 22807 5.613585 3.369298e-08 22807 2.575828e-05
## 22689 5.469138 7.712167e-08 22689 5.240846e-05
## 17348 -5.204884 2.888687e-07 17348 1.766721e-04

top_genes <- head(res_df$gene, 100)
mat_top_z <- t(scale(t(lcpm[top_genes, ])))

annotation_col <- data.frame(Class = sample_info$class)
rownames(annotation_col) <- sample_info$sample

pheatmap(
  mat_top_z,
  annotation_col = annotation_col,
  show_rownames = FALSE,
  width = 7,
  height = 9
)
```



```
top6 <- head(res_df$gene, 6)

df_bp <- data.frame(
  gene = rep(top6, each = ncol(lcpm)),
  sample = rep(colnames(lcpm), times = length(top6)),
  expr = as.vector(t(lcpm[top6, ]))
)
```

```
df_bp <- merge(df_bp, sample_info, by = "sample")

ggplot(df_bp, aes(x = class, y = expr, fill = class)) +
  geom_boxplot() +
  facet_wrap(~gene, scales = "free_y") +
  theme_minimal() +
  ggtitle("Boxplots top 6 genes")
```

